

India's Leap in Supercomputing: Innovating for Tomorrow

As India strides towards self-sufficiency in supercomputing, embracing this evolution isn't just an option—it is pivotal for global competitiveness and technological leadership.



Supercomputing architecture is undergoing a transformative shift globally, driven by the need for unprecedented computational power. High performance compute (HPC) machines stand out for their ability to perform complex calculations at speeds far beyond conventional computers. Nations

race to harness these capabilities for advancements in weather forecasting, drug discovery, artificial intelligence, and more. With significant strides in indigenous motherboard design and local manufacturing, coupled with ambitious initiatives like the National Supercomputer Mission, India's journey in supercomputing is the one to watch.

High performance compute (HPC) vs general compute

By definition, an HPC machine must be able to do 10x more computations than the fastest laptop or desktop available at that time.

An HPC machine must process multiple operations in parallel to achieve high computational capacity. A